Safety and Efficiency of Context-Aware Speed and Separation Monitoring in Human–Robot Ceiling-Panel Installation

Zenan Wu, Luke Siniakov, and Michael Magila

***Abstract*—Ceiling-panel installation is demanding overhead work that a collaborative robot could share by holding the panel while a worker fixes it. Fixed distance zones, today's safeguard, cannot tell whether the worker is approaching, working or moving away. This study asks whether a robot that recognises the task phase can keep the worker as safe as fixed-zone control while keeping the work efficient. We built a physical prototype in which a hidden Markov model recognises the task phase from motion capture and a separate predictor flags fast movement toward the robot, with a fixed rule that neither learned part can ever make the robot faster than the distance limit allows. Because every input was logged, the fixed-zone, reactive and predictive controllers could be compared on exactly the same recorded movement of every participant. All three stopped the robot every time a person entered the inner zone, but the predictive controller asked for the stop earlier for every participant, which gave far more protection when the measured distance was slightly wrong. The price was a little brief extra stopping that participants, who rated all three controllers as safe, did not notice. The hypothesis is therefore supported on safety and partly supported on efficiency. For industry, task awareness can be layered over an unchanged safety limit to buy margin against sensing error at little cost, while the larger productivity gain lies in a contact-capable mode while the worker is under the panel.**

***Index Terms*—Human–robot collaboration, speed and separation monitoring, hidden Markov model, construction robotics, motion capture, human factors.**

I. Introduction and Background

INSTALLING ceiling panels involves lifting, aligning and fixing panels overhead, work that is physically demanding and usually done by people alone or with lifting aids. Collaborative robots could take over the lifting and holding [1]; the arrangement adopted in this study is for a robot to hold the panel while the worker provides judgement and fine hand skills (Fig. 1a). The worker then moves around a robot that is carrying a panel, the panel enlarges the space the robot occupies, and unnecessary robot stops can break the flow of work.

Speed and separation monitoring (SSM) keeps enough distance between a person and a robot, and slows or stops the robot in time [2]–[4]. The simplest form uses fixed distance thresholds. These are easy to understand, but they treat a person walking toward the robot, standing still and walking away in the same way. Knowing the task phase could help the robot avoid unnecessary stops, as long as a wrongly recognised phase can never loosen the safety limit. Our controller therefore keeps three parts separate: task-phase recognition, motion prediction and the distance limit.

The aim is to make human–robot panel installation practical by protecting the worker without needless stops. This leads to the research question (RQ): can a controller that knows the task phase keep people as safe as the fixed-zone control used in practice today, while keeping the task efficient? Three supporting questions follow, answered in Section VII-A: (SQ1, safety and efficiency) on identical human movement, does task context change when and how often the robot is asked to stop; (SQ2, trust in the safety result) how robust is each controller to distance error, and how quickly does the robot stop; and (SQ3, acceptance) can participants tell the controllers apart. Our main hypothesis (H1) is that the context-aware predictive controller is as safe as fixed-zone control, the current industry approach, while maintaining comparable task efficiency. Its safety rests on the same distance limit: the learned parts are not allowed to relax it.

This work makes three contributions: (1) a physical ceiling-panel prototype in which task-phase recognition and intrusion prediction are kept separate, and both sit below a fixed speed limit; (2) an exact same-input comparison of the three controllers on the recorded movement of all 12 participants, including measured robot stopping and sensitivity to distance error; and (3) participant ratings linked to the controller that ran in each block. The novelty is the approach, not a new SSM formula: the controllers are compared on exactly the same human movement.

II. Related Work

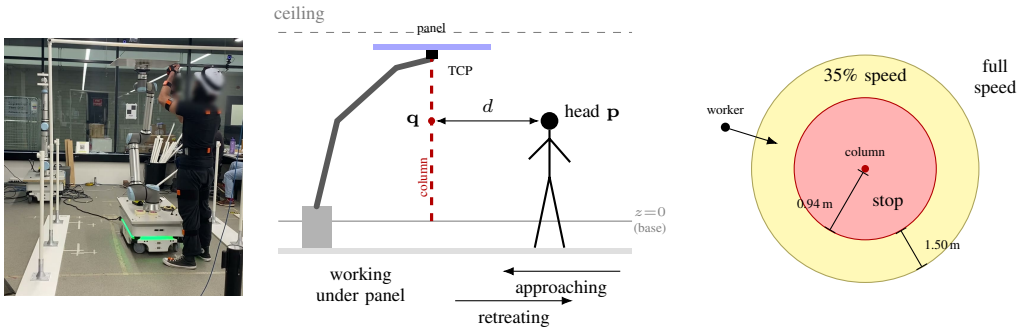
A. Human–Robot Collaboration in Construction

Construction robotics is moving from fully automated machines toward systems that combine human skill with robot strength [1], [9]. Reviews of human–robot collaboration (HRC) in construction list safety, sensing, communication and task sharing as the main barriers to use on site [10], [11]. Panel installation is a useful test case because the person and the robot work around the same object, and stopping too often directly slows the work.

B. Speed and Separation Monitoring

SSM needs more than a fixed safety radius around the robot. The required distance depends on how far the person

The authors are with the Department of Civil Engineering, Monash University, Melbourne, Australia. This work is a 2026 Final Year Project supervised by Yihai Fang.



(a) (b) (c)

Fig. 1. The application scenario. (a) Ceiling-panel installation in our lab: the robot holds the panel at ceiling height while the worker fixes it from below (frame from session video; face blurred). (b) Side view (schematic): the worker approaches, works under the panel and retreats; the robot is represented by a vertical column under the tool centre point (TCP), and d is measured from the head $\mathbf{p}$ to the nearest column point $\mathbf{q}$ (later system version). (c) Plan view, radii to scale: the fixed controller's zones around the column and the speed allowed in each.

and the robot move while the system senses, decides and brakes. It also includes margins for body parts that are not tracked and for sensor error [3], [4]. Byner et al. [5] set the speed limit from separation and direction of motion in a physical machine-tending cycle, and Karagiannis et al. [6] switch zone layouts with robot position using safety-rated hardware in a car-industry cell. Adaptive SSM and its effect on productivity have therefore already been studied, and they define the reactive baseline that task context must improve on.

Construction studies narrow the gap further. Xiao Lin et al. [7] combine worker, robot and material shapes in a digital twin that recalculates the protective distance as the scene changes, and test it in static and moving conflict scenarios. Their treatment of carried material matters for panel installation: once the robot carries a panel, a shape drawn around the robot alone no longer covers the space it occupies.

C. Activity Recognition and Collaboration

A hidden Markov model (HMM) tracks a state that changes over time but cannot be observed directly. It uses a transition matrix, which gives how likely the state is to change, and an emission model, which gives the measurements each state tends to produce [12]. HMMs have been used to recognise activities from wearable sensors and to model human–robot tasks [13], [14].

The closest construction study, by Xiao Lin et al. [8], combines a model of the work steps, robot behaviour learned from demonstration and local safety control in a virtual-reality wall-panel task with simulated sensors. Our study adds a physical ceiling-panel prototype with real robot data and participant questionnaires.

How smoothly people and robots work together is usually measured with timing, such as human idle time, robot idle time, time spent working in parallel and delay between actions [15]. Workload and trust questionnaires help explain results, but they cannot prove physical safety or replace measured robot response [16], [17].

Gaps addressed by this study. Three gaps follow from this review (Table I). First, adaptive SSM has been tested on physical robots mainly in manufacturing cells [5], [6], while the construction studies that model carried material or work steps used test scenarios or VR with simulated sensors [7], [8]. Second, in the studies reviewed each controller runs on its own trials, so differences in how people moved are mixed with differences between controllers. Third, recognition studies report accuracy, but not what a recognition or distance error does to the safety margin. Our research question, H1 and the same-input comparison (Section V-C) target these gaps.

III. System Design

A. Overview

The controller (Fig. 2) has two inputs: from motion capture, the position of the worker's head relative to the robot and the head's velocity in the room frame; and from the robot

TABLE I
Related methods and how they relate to this study.

Study	Method and test setting	What it means for this study
Byner et al. [5]	Speed limit set continuously from distance and direction of motion; tested in a physical machine-tending cycle.	Reactive speed adjustment is the baseline; phase recognition must add value on top of it.
Karagiannis et al. [6]	Safety zones switch with robot position; overhead safety camera and safety PLC in a car-industry cell.	A software speed limit is not the same as a system built from safety-rated parts.
X. Lin et al. [7]	Protective distance updated from worker, robot and material shapes; construction conflict scenarios.	Panel and body shapes need checking beyond whether a point-based distance is calculated consistently.
X. Lin et al. [8]	Learned work-step model and local safety control; VR wall-panel task with simulated sensors.	Task-aware safety control already exists; our focus is real robot data and participant experience.
This study	Real-time HMM task phase and motion prediction below a fixed speed limit; physical ceiling-panel prototype.	Robot data and questionnaires from a construction-style task with a person in the loop.

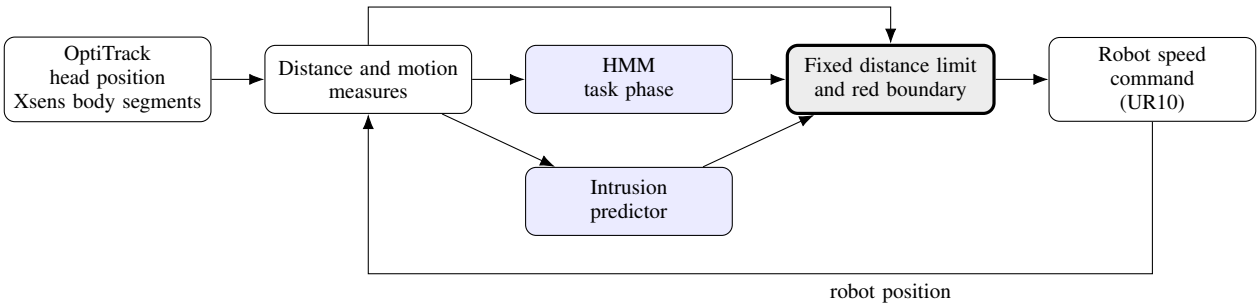


Fig. 2. How the prototype works. Motion capture and the robot's position give the distance and motion measures. The learned parts (blue) can only slow the robot further; they cannot raise its speed above the fixed distance limit (grey). Data collection for the experiment is shown separately in Fig. 4.

controller, the position of the robot's tool. Its output is a single number between 0 and 1: the fraction of full speed the robot may use. This number is sent to the robot's speed override, and a value of zero requests a pause.

B. Equipment and Sensing

The lab prototype (Fig. 3) uses a Universal Robots UR10 arm with a vacuum gripper that holds a lightweight stand-in for a ceiling panel. The arm lifts the panel to ceiling height, the participant walks in to work on it, and the arm then lowers it. Two separate systems track the participant.

Xsens MVN is a motion-capture suit. Seventeen small motion sensors, each with an accelerometer, gyroscope and magnetometer, are strapped to the body. Software combines their signals with a body model to give the position and orientation of 23 body segments at up to 240 Hz [18], [19]. Position is integrated, so it drifts, and its origin is set by a calibration pose rather than the room. Xsens provides the body segments for the later version's body-distance input.

OptiTrack is a camera-based motion-capture system. Infrared cameras around the cell (on the overhead frame in Fig. 3a) detect small reflective markers, and the Motive software locates each marker to about a millimetre in a fixed room frame [20]. A group of markers on the participant's cap is tracked as one object, giving the head position without drift but nothing about the rest of the body. It drives the phase model's distance and closing speed.

The two systems' estimates are not merged; both feed the same processing step in the same format. The robot controller reports the tool centre point (TCP), which is the position of the gripper face, and the speed of each of the six joints. The control loop runs at about 60 Hz, and the actual time between samples is logged.

C. Distance Measures and System Versions

The robot is represented by a vertical column, a simple stand-in for its shape (Fig. 1b). The column is a vertical line segment, with no radius, through the horizontal position of a robot reference point; it runs from the robot-base plane ($z = 0$) up to the height of that point. All positions are in the robot-base frame. For the tracked head position $\mathbf{p}$, the nearest point $\mathbf{q}$ on the column lies on the column at the head's height, limited to the column's height range. The controller calculates

$$d = \|\mathbf{q} - \mathbf{p}\|_2, \qquad v_{\mathrm{proj}} = \mathbf{v}^{\mathsf{T}}\mathbf{u}, \tag{1}$$

TABLE II
SYMBOLS USED IN THE CONTROLLER EQUATIONS.

Symbol	Meaning
d	Distance from tracked point to robot (m)
v_{proj}, a_{proj}	Speed (m/s) and acceleration (m/s^2) toward the robot
h_t	How directly the movement points at the robot
z_t	Task phase at time t (approaching, working, retreating)
$w_{ik}, \boldsymbol{\mu}_{ik}, \boldsymbol{\sigma}^2_{ik}$	Weight, mean and variance of component k in phase i
r_{red}	Radius of the inner (red) boundary (m)
τ	Predicted time to reach the red boundary (s)
S	Protective separation distance (m)
T	Assumed time to react and stop (s)
C	Margin for body parts not tracked (m)
S_a	Margin for sensor error (m)
$u_{\mathrm{env}}, u_{\mathrm{context}}, u_{\mathrm{cmd}}$	Speed fractions (0–1) from the distance limit, from the learned parts, and sent to the robot

where d is the head-to-robot distance in metres, $\mathbf{v}$ is the head velocity in the room frame (the least-squares slope of the last five samples; the robot's own motion is not subtracted), $\mathbf{u}$ is the unit vector pointing from the head toward the robot, and v_{proj} is the closing speed in m/s. A positive v_{proj} means the person is moving toward the robot. The heading alignment h_t, used below, measures how directly the person's movement points at the robot; it is not the direction the body faces. Table II lists all symbols.

Data were recorded with two system versions that place the column differently. In the earlier version the reference point was fixed on the robot base's vertical axis, 2.2 m high, so the column did not follow the arm. In the later version it is the live TCP, read at every step and used only if less than 0.25 s old, so the column moves with the gripper and panel. The later version also streamed data automatically rather than on operator confirmation, and for control it used the 23 Xsens body segments, placed at the OptiTrack cap and taking the smallest distance and largest closing speed. The phase model used head measures in both. The versions are reported separately (Section V-B), not as an experimental condition.

D. Task Phase and Intrusion Prediction

The model recognises three task phases: approaching, working and retreating. These labels describe what the person is



(a) (b) (c)

Fig. 3. Lab equipment. (a) The cell: the UR10 arm holding the panel stand-in at ceiling height, OptiTrack cameras on the overhead frame, and the operator desk. (b) A participant wearing the Xsens suit and OptiTrack cap, seen from behind while walking toward the robot. (c) The working phase: the participant works under the raised panel. Faces are blurred.

doing, whereas a hazard is about being exposed to harm, and a person can trip or move quickly toward the robot during any phase. Treating "hazard" as a fourth phase would therefore mix up activity and danger. The HMM estimates only the task phase, and fast movement toward the robot is detected separately by the intrusion predictor below, so the distance limit can be tested on its own whatever the HMM decides. At each time step it uses four measures, $\mathbf{x}_t = [d_t, v_{\mathrm{proj},t}, \|\mathbf{v}_t\|, h_t]^{\mathsf{T}}$: distance, closing speed, overall speed and heading alignment. For each phase, the typical values of these measures are described by a mix of three Gaussian (bell-shaped) distributions:

$$p(\mathbf{x}_t \mid z_t = i) = \sum_{k=1}^{3} w_{ik} \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_{ik}, \mathrm{diag}(\boldsymbol{\sigma}_{ik}^2)), \qquad (2)$$

where z_t is the phase at time t, w_{ik} is the weight of component k in phase i (the three weights add to one), and $\boldsymbol{\mu}_{ik}$ and $\boldsymbol{\sigma}_{ik}^2$ are that component's mean and variance for each measure. In real time the model can use only present and past data (the causal filter). For offline checking, the Viterbi method can also use later data, so results for the two are reported separately. Because the labelled phases are long and the model expects phases to persist, it changes phase slowly and may react late to short movements.

The intrusion predictor works independently of the HMM. It solves

$$d - r_{\mathrm{red}} - v_{\mathrm{proj}}\tau - \tfrac{1}{2} a_{\mathrm{proj}} \tau^2 = 0 \qquad (3)$$

for τ, the time the person would take to reach the inner (red) boundary of radius r_{red} from the current distance d if they kept their current closing speed v_{proj} and acceleration a_{proj}. An intrusion is flagged when a risk score that rises as τ falls below 0.5 s passes its threshold (equivalent to $\tau \lesssim 0.58$ s) while the closing speed is at least 0.6 m/s. a_{proj} is the slope of the last three closing speeds, a derivative of a noisy derivative that reached about ± 30 m/s^2 unclamped during development, far beyond a walking person. It is therefore capped at ± 4 m/s^2 (about $0.4\,g$), and the 0.6-m/s gate stops sway near the boundary counting as an approach. Both are engineering settings chosen in simulation to suppress noise, not measured or standard values. In the later version the body-segment control input sets $a_{\mathrm{proj}} = 0$, so the cap had no effect there. The predictor detects movement toward the boundary, not intention.

E. Controllers and Speed Command

Three controllers were compared. The *fixed* controller uses two zones around the column (Fig. 1c): inside the red radius it requests a stop, inside the yellow radius it limits the robot to 35% of full speed, and outside both it allows full speed. Leaving a zone requires clearing its boundary by 0.05 m, so the command does not flicker at the edge. The *reactive* controller also takes closing speed into account, using a simplified separation limit. The *predictive* controller adds phase-based caution and the intrusion predictor on top of the reactive limit. Its protective separation distance is

$$S = \max(0, v_{\mathrm{proj}})T + C + S_a, \qquad (4)$$

where S is the distance the controller tries to keep; T is the assumed time for the system to react and the robot to stop, so $v_{\mathrm{proj}}T$ is how much closer the person gets in that time; C is a margin for body parts that the tracked point does not cover; and S_a is a margin for sensor error. This is the ISO/TS 15066 separation formula without the term for the robot's own movement [3]. We used $T = 0.4$ s, $C = 0.2$ m and $S_a = 0.1$ m. Beyond S, the allowed speed rises gradually over 0.3 m. The red radius, 0.94 m, is the same formula evaluated at an assumed approach speed of 1.6 m/s ($1.6T + C + S_a$), and the yellow radius is 1.6 times that (1.50 m). These values were chosen for the lab cell; they are not measured stopping times or sensor errors of an installed system.

The speed sent to the robot is

$$u_{\mathrm{cmd}} = \min(u_{\mathrm{env}}, u_{\mathrm{context}}), \qquad 0 \le u_{\mathrm{cmd}} \le 1, \qquad (5)$$

where u_{env} is the speed fraction allowed by the reactive limit, u_{context} is the fraction suggested by the phase model and intrusion predictor, and u_{cmd} is the fraction sent to the robot. Taking the smaller of the two means the learned parts can only lower the speed, never raise it. Inside the red radius the robot is always asked to stop. If the head position, body data or robot position are missing or too old (more than 150 ms for OptiTrack or 0.25 s for the robot position), every controller requests a stop ($u_{\mathrm{cmd}} = 0$) before its own rules are applied. The command is sent through the robot's speed override and pause function. This is a research controller: its stop is not a safety-rated protective stop, and (5) limits the software request rather than certifying the installed system (Section VII-E).

The HMM's phase caution can only cap the speed at the reduced level, so it never stops the robot on its own.

TABLE III
LEAVE-ONE-OPERATOR-OUT PHASE-RECOGNITION VALIDATION ON THE DEVELOPMENT SET. VITERBI DECODING USES THE FULL SEQUENCE; THE CAUSAL FILTER USES PRESENT AND PAST OBSERVATIONS.

Measure	Value
Development participants	3
Accepted trials	32
Labelled frames	90487
Offline Viterbi accuracy	0.821
Offline balanced accuracy	0.757
Causal-filter accuracy	0.794
Causal-filter balanced accuracy	0.725
Approaching recall, offline / causal	0.693 / 0.684
Working recall, offline / causal	0.923 / 0.905
Retreating recall, offline / causal	0.654 / 0.586
Offline fold P03 accuracy	0.812
Offline fold P04 accuracy	0.817
Offline fold P05 accuracy	0.832

The intrusion predictor stops once the risk has exceeded its threshold for two consecutive samples and releases as soon as it falls back below the same threshold, with no hysteresis.

IV. MODEL DEVELOPMENT AND VALIDATION

We trained the phase model before the participant study, using 32 labelled trials in which the three team members acted as operators (study IDs P03, eight trials; P04, 12; P05, 12). These contain 90,487 frames: 17,267 approaching, 53,609 working and 19,611 retreating. Labels were recorded live: the experimenter pressed a key at each real phase change to advance the dashboard's fixed task sequence, and each frame carries the current phase. *Approaching* runs from the start cue until the participant reaches the gripper with the panel or stands under the raised panel; *working* runs until the panel is gripped or, in the second cycle, a brief hazard movement is cued; *retreating* runs until the robot's lift or lowering is confirmed. Frames during robot motion and the hazard movement are excluded. All 32 trials follow the scripted order, but the labels were not reviewed afterwards or checked by a second labeller. We validated it by leaving one operator out at a time: the model was trained on two operators and tested on the third.

Two measures are reported. Accuracy is the share of frames whose predicted phase matches the label. Because most frames are "working", a model that favours that phase scores well on accuracy. Balanced accuracy averages the recall of the three phases, so each phase counts equally. Offline accuracy is 0.821 and real-time (causal) accuracy is 0.794 (Table III). Balanced accuracy is lower, especially for the causal filter (0.725), and recall is highest for the working phase. Because the controller runs online, the causal recalls are the relevant ones: 0.684 for approaching, 0.905 for working and 0.586 for retreating. Retreating loses most, as the filter is slow to leave the working phase. The small number of operators, the correlation between frames in the same trial, and the reuse of the same folds during development limit how far these values generalise. The 12 participants were never used in training, so they form a further held-out test. Against the planned-cue phase labels, the real-time model was correct a median of 69% of the time per person (range 44–80%; balanced accuracy 65%). The labels mark when a cue was given rather than when the movement began, so this is a lower bound. Phase errors change only the yellow-zone speed limit, not stops.

V. EXPERIMENTAL METHOD

A. Task and Experimental Design

The protocol has three blocks, A, B and C. The fixed, reactive and predictive controllers were assigned to the blocks in one of six rotated orders. Each block contains one clean trial, one distractor trial and one rapid-movement trial (Fig. 4). Each trial follows the panel sequence: walk in and load the panel, step back before the lift, work under the raised panel, and step back before it is lowered and unloaded. The trial types differ only during the lift, while the participant waits at the start marker. In a *clean* trial there is no cue. In a *distractor* trial, on the operator's "now", the participant makes one brisk sideways movement that does not close on the robot and returns. In a *rapid-movement* trial they make one brief move toward the robot along a marked direction and return at once (not a fall, trip or lunge). Both movements were demonstrated and practised first. A cue window marks the instruction, not the start of the movement.

B. Participants and Analysed Records

The study ran under Monash University ethics approval (project 51642). Twelve people took part. Some sessions were restarted under a new P-number, so we checked identity from the data before analysis. Each analysed study code is one continuous sitting of 25 to 35 minutes that contains a full rotated sequence of the three controllers, separated from the next sitting by about 30 minutes. The Xsens body profile was sometimes reused between people (two pairs of codes have identical segment lengths), so it cannot identify a person; the height of the tracked helmet, measured independently by OptiTrack, separates those codes (for example 1.75 against 1.81 m, with a spread of about ± 0.01 m within a code). One person was recorded under two codes on 17 September, confirmed by identical intake answers; these are merged. Session video was not used for this check. In this paper a code such as P17 is a study ID, and T05 is a trial within it. Trials from P13–P22 used the earlier system version and those from P31 onward the later one (Section III).

Eleven people have eight or nine analysed trials each (97 trials; two planned slots had no completed trial). The twelfth person has eight trials (three reactive and five predictive, with no fixed-zone block) that were excluded from the earlier analysis for phase order and a lower capture rate (about 26 Hz); they are included here because the same-input replay (Section V-C) gives all three controllers on their recorded movement. In total there are 105 trials. Eleven people gave 31 block questionnaires (two are missing the last block); the twelfth completed only the intake form. Pilot, rehearsal and team sessions are excluded. No missing values were filled in.

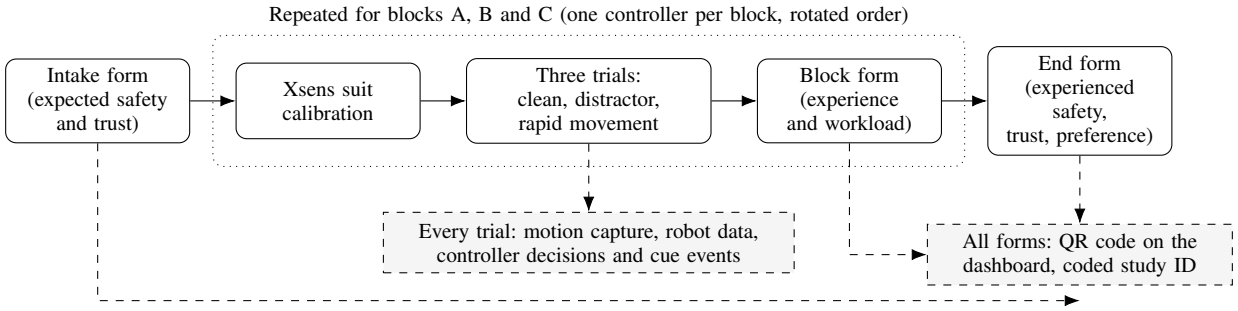


Fig. 4. One participant session. Solid arrows show the order of steps; dashed arrows show the data collected. The three controllers were assigned to blocks A, B and C in one of six rotated orders.

C. Robot Data Processing

The selection above gave 97 trials and 494,075 samples. File fingerprints, row counts and time totals were rechecked independently after the analysis. For time totals, each sample's value is held until the next sample, but only when the gap to the next sample is between 0 and 0.25 s. Longer gaps are reported separately and left out of both the total and the time base. A robot-data sample counts as fresh when the packet is available, all six joint speeds are valid and the data are less than 0.25 s old. The robot counts as moving when its fastest joint exceeds 0.005 rad/s; counts at 0.001 and 0.01 rad/s are kept as a check on this threshold.

To check the head distance, we recalculated it from the same sample's head position and the robot's live tool position. This checks internal consistency, not physical calibration. For each cue window we report the smallest distance and the largest closing speed, which need not occur at the same moment. Stop time includes intentional holds in the task, so it is not a count of false stops.

Same-input replay. The controllers are fixed rules applied to logged inputs (distance, closing speed and acceleration, and body features), so each can be re-run on any recorded trial. We replayed all three controllers on the recorded inputs of every trial, with the same fail-safe holds the live system applied to all of them. Replay used the exact code and model version that ran: the first participant's sitting used an earlier version of the predictive controller's yellow-zone rule, and this was reproduced. The replayed command matched the logged robot command in every sample of all 105 trials, and, in the later trials where the system also logged the other two controllers' requests, it matched those too. This gives a paired comparison of the three controllers on identical human movement, free of differences in person, order, learning and fatigue. It is open loop: it shows what each controller would have requested, not how the person or robot would then have moved.

Measures. A stop is *unnecessary* if the person is outside the red boundary and does not enter it within the next H s ($H = 1$ s; 0.5 and 2 s give the same ranking). For each entry into the red boundary, *lead* is how long a stop had already been requested. To test sensitivity to distance error, entries and coverage are recomputed with the red boundary moved by ± 5 and ± 10 cm while the controllers are unchanged. Robot stopping time is measured from a stop request made while the tool was moving to the first sample at which the measured tool speed is below 0.005 m/s. Stop requests are also classed as brief (under 0.5 s) or sustained.

Statistics. The person is the unit of analysis. Quantities are totalled per person and divided by that person's observed time. Paired differences are tested with the Wilcoxon signed-rank test, with 95% confidence intervals from a bootstrap over people.

D. Questionnaire Inclusion and Scoring

Pilot, link-test and qualification submissions were excluded. This left 18 intake forms, 31 block forms and 11 end forms, with 11, 11 and 9 block forms for A, B and C and no duplicates.

The block questionnaire uses five-point ratings. Three items run between two opposite words: anxious to relaxed, agitated to calm, and quiescent to surprised. Four items ask how much the participant agrees that they were confident the robot would stop, had to watch it constantly, felt it stopped more than needed, and found its speed changes made sense. Three items rate mental demand, physical demand and frustration. The workload items are adapted from NASA-TLX [16] and the confidence and monitoring items from Jian et al. [17]. Neither set is the full validated questionnaire, so each item is analysed on its own and no combined score is formed.

Every trial record states its block and the controller that ran, so each block questionnaire is linked to a controller. Items are compared between controllers with the Friedman test (people with all three blocks) and paired Wilcoxon tests, and against block position (first, second, third) to separate controller from order. End-form choices are mapped to controllers and compared with chance. Whether felt unnecessary stopping follows measured unnecessary stopping is tested with a within-person (repeated-measures) correlation. The intake and end forms ask about expected and then experienced safety and trust, from 1 (not at all) to 5 (completely).

VI. Results

A. Robot Data

The 97 trials of the eleven people with full sessions are summarised by the controller that actually ran (Table IV); these logged totals mix person, order and system version, and Section VI-B gives the paired comparison. Median stop time,

TABLE IV
RECORDED BEHAVIOUR BY CONTROLLER (97 TRIALS). BRIEF STOP REQUESTS LAST LESS THAN 0.5 S.

	Fixed	Reactive	Predictive
Trials	32	32	33
Clean / distractor / rapid	10/11/11	10/11/11	11/11/11
Median trial duration (s)	112.5	113.6	109.0
Median closest distance (m)	0.08	0.06	0.08
Time inside red boundary (%)	28.5	31.3	32.0
Stop requested inside red (%)	100	100	100
Median stop time (%)	29.7	33.6	36.7
All stop requests	659	888	1417
Brief: sensing dropout hold	450	414	326
Brief: intrusion predictor	0	0	805
Brief: other rules	43	293	69
Sustained: red boundary	164	156	101
Sustained: intrusion predictor	0	0	104
Sustained: other rules	2	25	12
Sustained stops per minute	2.7	3.1	3.6

Totals are descriptive, not controller-effect estimates: order, participant identity and system version are not jointly controlled. Stop time includes intentional task holds. P13 A1 and P22 A3 have no completed capture.

which includes intentional task holds, is 29.7%, 33.6% and 36.7% for fixed, reactive and predictive records.

The safety indicators were similar across controllers (Table IV). Whenever the measured distance was inside the 0.94-m red boundary, all three controllers requested a stop 100% of the time. The median closest distance per trial was 0.08, 0.06 and 0.08 m, and people spent 28.5%, 31.3% and 32.0% of observed time inside the red boundary, mostly during the loading, working and unloading approaches the task requires. Median trial duration was 112.5, 113.6 and 109.0 s, so predictive trials were not slower overall, although the robot spent more of that time stopped.

B. Same-Input Comparison

Table V gives the replay of all three controllers on the identical movement of all 12 people. On this movement every controller had requested a stop at all 970 entries into the red boundary, and none ever requested motion inside it. The predictive controller started stopping earlier: its lead at each entry was 0.107 s longer than fixed-zone control (95% CI 0.076 to 0.139) and 0.096 s longer than reactive control, for all 12 people ($p = 0.0005$). It also stopped more without need: 0.90 s per minute more than fixed-zone control (95% CI 0.36 to 1.51) and 0.76 s more than reactive control, again for all 12 people ($p = 0.0005$). Most of the extra stop requests are brief switching. Unnecessary stopping is small for every controller: 94 to 98% of stopped time is followed by an entry into the red boundary within 1 s, because the task keeps the worker inside the protective distance for long periods. *For SQ1:* task context made the robot stop sooner without missing any entry, at a small cost in brief extra stops.

Distance error. The 100% coverage holds at the measured distance because all controllers share the red boundary. The reused body profiles put modelled body size off by up to about 5 cm. If the true distance were 5 cm smaller than measured, a stop would already have been requested at 67% of entries under predictive control, against 42% under reactive and 31% under fixed-zone control (Table V). The predictive advantage held for all 12 people, and the result is the same at 10 cm. Early stopping therefore turns a small lead time into a large margin against measurement error. *For SQ2:* the safety result does not depend on the distance being exact, and the predictive controller is the most tolerant of error.

TABLE V
SAME-INPUT COMPARISON: ALL THREE CONTROLLERS REPLAYED ON THE IDENTICAL RECORDED MOVEMENT OF ALL 12 PARTICIPANTS (105 TRIALS, 206 MIN, 970 ENTRIES INTO THE RED BOUNDARY). RATES ARE PER MINUTE; UNNECESSARY STOPS ARE STOPS WITH NO ENTRY INTO THE RED BOUNDARY WITHIN 1 S; LEAD IS HOW LONG A STOP HAD ALREADY BEEN REQUESTED WHEN THE PERSON ENTERED THE RED BOUNDARY.

	Stop	Brief	Unneces-	Lead	Entries covered (%)	
Controller	(s/min)	stops/min	sary (s/min)	(s)	measured	+5 cm error
Fixed zone	18.5	2.2	0.30	0.69	100	31
Reactive	18.8	6.7	0.44	0.70	100	42
Predictive	20.3	15.3	1.20	0.80	100	67

Robot stopping. The tool reached standstill a median of 0.156 s after a stop request (95th percentile 0.188 s, maximum 0.234 s; 226 requests made while the tool was moving). The tool moved inside the red boundary only while braking from a stop already requested, and never closer than 0.54 m under program motion. In 5 of the 12 people the held arm was briefly pushed a few millimetres while the person worked under the panel.

C. Distance Inputs and Cued Movements

In the early records, the median difference between the recorded head distance and the same-sample recalculation (95th percentile per trial) is 0.481 m (Fig. 5a). In the later records it is 0.000 m at the reported precision. The early difference arises because the recorded distance was measured to the column on the robot-base axis, whereas the recalculation uses the column under the live TCP, so it grows as the arm carries the panel away from the base axis. The early inputs therefore describe the controller's view, not the true head-to-panel clearance; the later zero shows arithmetic consistency, not physical calibration.

The wide spread and overlap of rapid and distractor cue windows (Fig. 5b) show that the scripted cue name does not decide the movement that was actually measured, so the measured inputs, not the cue name, guide interpretation. Figure 6 shows a typical later record. Because the cue marks the instruction rather than the start of the movement, the delay between cue and robot response is not reported as a protective response time.

D. Reported Experience

Block responses lean toward relaxed and calm, confidence that the robot would stop, and low demand and frustration (Table VI). Linked to the controller that ran in each block, no item differed between controllers (Friedman $p \geq 0.11$ for all ten items, 11 people, 31 blocks). Feeling relaxed rose with block order instead, from a mean of 3.78 in the first block to 4.67 in the third ($p = 0.011$), whatever the controller.

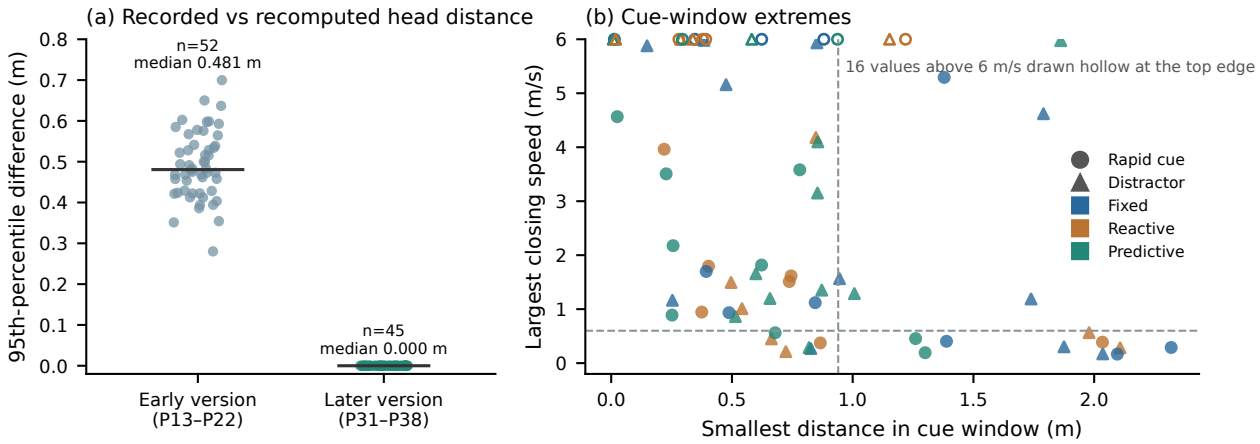


Fig. 5. Measurement checks, one mark per trial. (a) 95th-percentile difference between the recorded head distance and a recalculation from the same sample, by system version. (b) For each distractor or rapid cue window, the smallest distance (horizontal) against the largest closing speed (vertical). Dashed lines show the 0.94-m inner boundary and the 0.6-m/s closing threshold. Values above 6 m/s are physically implausible input spikes and are drawn hollow at the top edge. The two extremes need not occur at the same moment.

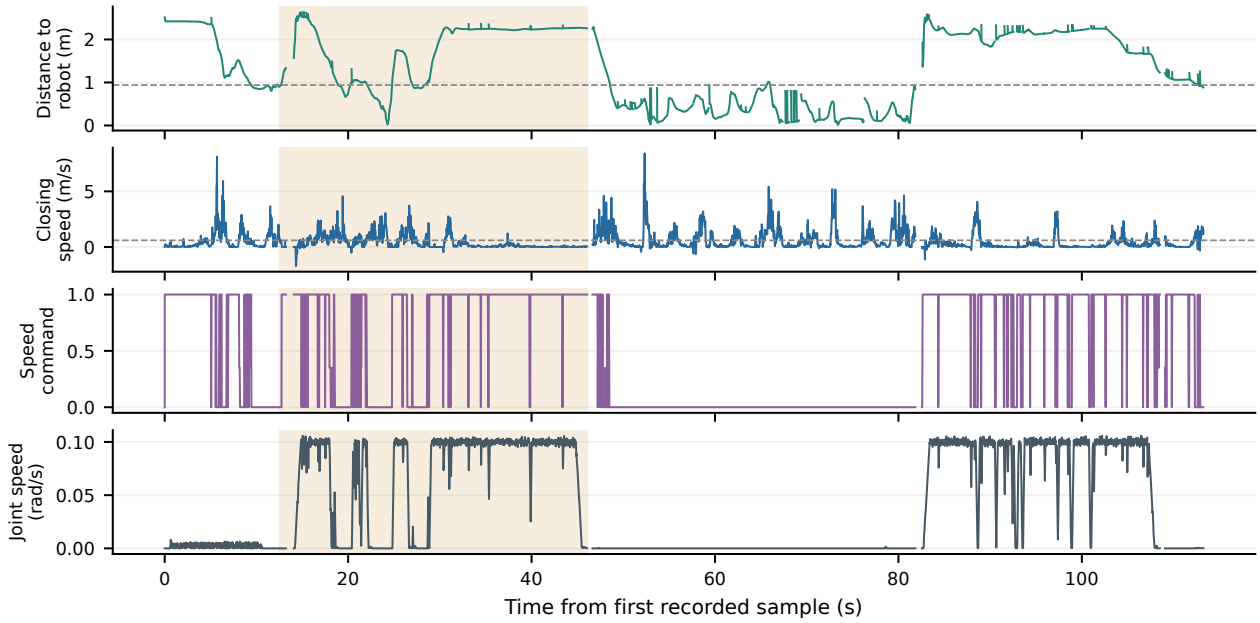


Fig. 6. A typical later trial (P31 T10, predictive controller): distance to the robot, closing speed, speed command and fastest joint speed. Shading marks the cue window; dashed lines are the configured thresholds; gaps over 250 ms break the lines. Note the rapid switching of the speed command.

Agreement that the robot "stopped more than it needed to" did not follow the measured unnecessary stopping in the same block (within-person $r = 0.17$, $p = 0.47$).

Across the 11 matched records, the mean safety rating was 3.82/5 before and 4.45/5 after the session, and mean trust was 4.00 before and 4.18 after (Table VII).

In the end forms, the block chosen as safest was most often the last one (C, seven times), but mapped to controllers the choices were fixed zone 3, reactive 5 and predictive 3, which is what chance would give. The preference for the last block is therefore an order effect, consistent with the rise in relaxation. *For SQ3:* people felt safe with every controller, so the extra caution cost nothing in perceived comfort, but ratings cannot rank the controllers. Four participants noticed changes between blocks and seven did not. The median rating for how much the suit restricted movement was two of five, and no end form reported risk or discomfort.

VII. Discussion

A. Answering the Research Question

The supporting questions combine into one answer. On identical movement the task-aware controller missed no entry and stopped earlier (SQ1); that lead holds when the distance is a few centimetres wrong, the realistic failure (SQ2); and workers accepted it as readily as fixed zones (SQ3). H1 is therefore supported on safety and only partly on efficiency: trials took about as long, but the robot paused slightly more. Against the aim, it protects the worker better without slowing the work, but does not remove the stops that interrupt it

TABLE VI
BLOCK QUESTIONNAIRE ITEMS: MEDIAN [FIRST, THIRD QUARTILE]. A, B AND C ARE REPORTED BLOCK LABELS, NOT POOLED CONTROLLER CONDITIONS. ALL RATINGS USE 1–5.

Item	A ($n = 11$)	B ($n = 11$)	C ($n = 9$)
Anxious to relaxed	4 [3.5, 4.5]	5 [4.0, 5.0]	5 [4.0, 5.0]
Agitated to calm	4 [4.0, 5.0]	5 [4.0, 5.0]	5 [4.0, 5.0]
Quiescent to surprised	3 [1.0, 3.0]	3 [1.0, 4.0]	1 [1.0, 3.0]
Confident robot would stop	5 [4.0, 5.0]	5 [4.0, 5.0]	5 [4.0, 5.0]
Had to watch constantly	2 [1.0, 3.0]	2 [1.0, 3.0]	1 [1.0, 3.0]
Stopped more than needed	3 [2.0, 4.0]	2 [1.0, 3.5]	2 [2.0, 3.0]
Speed changes made sense	5 [3.5, 5.0]	5 [4.0, 5.0]	4 [4.0, 5.0]
Mental demand	1 [1.0, 3.0]	1 [1.0, 2.0]	1 [1.0, 2.0]
Physical demand	1 [1.0, 3.0]	1 [1.0, 2.5]	1 [1.0, 3.0]
Frustration	1 [1.0, 3.0]	1 [1.0, 2.0]	1 [1.0, 1.0]

Higher means the right-hand semantic anchor for the first three items, stronger agreement for the next four, and greater demand/frustration for the final three. Responses are clustered within 11 study IDs; individual-level linkage is incomplete. No trust, safety or workload composite is used.

TABLE VII
PAIRED GLOBAL RATINGS FOR 11 CODES WITH INTAKE AND END RESPONSES. FIVE-POINT ITEMS; DESCRIPTIVE CODE-LEVEL CHANGES ONLY.

Item	Before	After	Δ	Up/same/down
Felt safe	3.82	4.45	0.64	8/1/2
Trust in distance	4.00	4.18	0.18	5/3/3

Before/after wording changes from expected to experienced safety/trust. Means summarize coded responses; neither causation nor an interval-scale measurement model is assumed.

(Section VII-D). For construction firms and integrators the lesson is architectural: learned task recognition can sit on an unchanged, standard distance limit, so an assessor still validates only that limit while the learned parts add caution where sensing is weakest.

B. What Prediction Buys

Because the learned parts sit below the distance limit, the predictive controller can never request a higher speed than the reactive one on the same inputs; any benefit must come from when it intervenes. The same-input replay measures exactly this. Prediction moved the stop request earlier for every participant, and under a few centimetres of distance error that earlier request covered roughly twice as many boundary entries as fixed-zone control. The value of prediction in this task is therefore robustness to measurement error, which matters because body distance is the least certain input in the system. Where Byner et al. [5] used direction-aware limits to gain productivity, task context here bought margin rather than speed, and the result supports X. Lin et al.'s [7] view that body and material shape decide whether the protective distance is adequate.

The cost is brief switching. The intrusion predictor stops the robot after two samples above its threshold and releases as soon as the risk drops, so a noisy closing-speed estimate near the threshold makes the command flip between stop and run. This added under one second of unnecessary stopping per minute, and participants did not notice it: no rating differed between controllers, which shows why ratings cannot replace measured robot response [17]. Because the switching is driven by real tracking noise, a test with simulated sensors such as [8] would not reveal it. A release delay or hysteresis on the predictor should remove most of the switching without touching the fixed distance limit, but a simple minimum stop time is not a free fix, because it lengthens every stop.

C. The Task Limits the Gain

Under all three controllers the robot was stopped for about a third of each trial, and almost all of that stopping was followed by the worker entering the red boundary within a second. The panel task places the worker inside the protective distance while loading, working under the panel and unloading, and several participants pushed the held panel. The productivity limit therefore comes from using SSM during close-contact phases, not from the choice of SSM controller. Switching to a contact-capable mode (hand guiding or power and force limiting [3]) by task phase, much as Karagiannis et al. [6] switch zone layouts by robot position, is the larger opportunity, and the system design already supports it.

D. Safety and Efficiency

Research on adaptive SSM motivates measuring robot availability and cycle performance [5], [6]. The measurable part of the reaction chain fits the assumed $T = 0.4$ s: Xsens data were at most 0.078 s old in 95% of control steps, a step took at most 0.047 s, and the robot stopped within 0.234 s, giving about 0.31 s at the 95th percentile. The rarest cases reach about 0.44 s, and the processing delay inside the motion-capture devices is not visible in the logs, so a larger T (for example 0.5 s) is advisable. Stop time includes intentional holds and capture time includes operator actions, so neither directly measures efficiency or fluency [15]. The command used the ordinary robot speed override rather than a safety-rated stop, so none of these results support a certified safety claim.

E. Limitations

The sample is small (12 people) and was recruited for convenience; one participant lacks the fixed-zone block and the questionnaires after the intake. The system version changed during collection: the earlier six participants were measured

with a head-only distance to a column fixed on the robot-base axis, so their boundary entries and lead times refer to the controller's distance rather than the true clearance to the moving panel; the later six used a whole-body distance to the live TCP, and the first participant used an earlier version of the predictive yellow-zone rule. The Xsens body profile was reused between some people, which introduces up to about 5 cm of distance error; Section VI-B reports how much this matters. The same-input comparison is open loop, so it does not show how people would have moved under a different controller. The unnecessary-stop measure uses hindsight over a 1-s window and the red boundary as the reference, and the phase labels mark cues rather than observed movement. Cued movements, operator pacing, learning and fatigue may affect behaviour and ratings; the rotated block order controls for order but not for these.

Lessons for similar studies: freeze the system version before the first participant; log the geometric reference beside each distance, which would have exposed the base-axis column at once; and log every input and every controller's request, which made the exact replay possible and verifiable.

F. Future Work

A confirmatory session should apply these lessons and run the predictive controller with a release delay on the intrusion predictor, a reaction time of at least 0.5 s and freshly measured body profiles for each person. It should also switch to a contact-capable mode while the worker is under the panel, since that is where most of the stopping occurs.

VIII. Conclusion

This study asked whether a robot that knows the task phase can keep a ceiling-panel installer as safe as fixed-zone control while keeping the work efficient. An exact replay of all three controllers on the recorded movement of 12 participants gave a clear answer on safety: every controller stopped for every close approach, and the task-aware controller stopped earlier for every participant, keeping protection high even when the measured distance was a few centimetres wrong. A little brief extra stopping, unnoticed by participants and removable with a release delay, left efficiency comparable but not improved. For industry, task awareness is a low-risk way to add margin on top of a standard safety limit; the larger productivity gain lies in switching to a contact-capable mode while the worker is under the panel.

Data Availability

Analysis code and derived tables accompany the manuscript; participant-level data require controlled access. Nothing was imputed or simulated.

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AI tools assisted with data-processing and analysis code, figure and table generation, and drafting and editing of the manuscript text.

References

[1] M. Zhang, R. Xu, H. Wu, J. Pan, and X. Luo, "Human–robot collaboration for on-site construction," *Automation in Construction*, vol. 150, p. 104812, 2023.

[2] International Organization for Standardization, "ISO 10218-2:2025 robotics—safety requirements—part 2: Industrial robot applications and robot cells," International Standard, 2025. [Online]. Available: https://www.iso.org/standard/73934.html

[3] ——, "ISO/TS 15066:2016 robots and robotic devices—collaborative robots," Technical Specification, 2016. [Online]. Available: https://www.iso.org/standard/62996.html

[4] J. A. Marvel and R. Norcross, "Implementing speed and separation monitoring in collaborative robot workcells," *Robotics and Computer-Integrated Manufacturing*, vol. 44, pp. 144–155, 2017.

[5] C. Byner, B. Matthias, and H. Ding, "Dynamic speed and separation monitoring for collaborative robot applications—concepts and performance," *Robotics and Computer-Integrated Manufacturing*, vol. 58, pp. 239–252, 2019.

[6] P. Karagiannis, N. Kousi, G. Michalos, K. Dimoulas, K. Mparis, D. Dimosthenopoulos, Ö. Tokçalar, T. Guasch, G. P. Gerio, and S. Makris, "Adaptive speed and separation monitoring based on switching of safety zones for effective human robot collaboration," *Robotics and Computer-Integrated Manufacturing*, vol. 77, p. 102361, 2022.

[7] X. Lin, Z. Guo, X. Jin, and H. Guo, "Digital twin-enabled safety monitoring system for seamless worker–robot collaboration in construction," *Automation in Construction*, vol. 174, p. 106147, 2025.

[8] X. Lin, Z. Guo, H. Guo, and X. Jin, "Safety-aware behavioral decision-making for construction robots: Semantic-driven hybrid framework for worker–robot collaboration," *Automation in Construction*, vol. 189, p. 107067, 2026.

[9] T. Bock, "The future of construction automation: Technological disruption and the upcoming ubiquity of robotics," *Automation in Construction*, vol. 59, pp. 113–121, 2015.

[10] H.-H. Wei, Y. Zhang, X. Sun, J. Chen, and S. Li, "Intelligent robots and human–robot collaboration in the construction industry: A review," *Journal of Intelligent Construction*, vol. 1, no. 1, p. 9180002, 2023.

[11] P. Lin, N. Zeng, Q. Li, and K. Nübel, "A review of human–robot collaboration safety in construction," *Systems*, vol. 13, no. 10, p. 856, 2025.

[12] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.

[13] N. Manouchehri and N. Bouguila, "Human activity recognition with an HMM-based generative model," *Sensors*, vol. 23, no. 3, p. 1390, 2023.

[14] J. Berg, T. Reckordt, C. Richter, and G. Reinhart, "Action recognition in assembly for human–robot-cooperation using hidden Markov models," *Procedia CIRP*, vol. 76, pp. 205–210, 2018.

[15] G. Hoffman, "Evaluating fluency in human–robot collaboration," *IEEE Transactions on Human-Machine Systems*, vol. 49, no. 3, pp. 209–218, 2019.

[16] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (task load index): Results of empirical and theoretical research," in *Human Mental Workload*, P. A. Hancock and N. Meshkati, Eds. Amsterdam: North-Holland, 1988, pp. 139–183.

[17] J.-Y. Jian, A. M. Bisantz, and C. G. Drury, "Foundations for an empirically determined scale of trust in automated systems," *International Journal of Cognitive Ergonomics*, vol. 4, no. 1, pp. 53–71, 2000.

[18] D. Roetenberg, H. Luinge, and P. Slycke, "Xsens MVN: Full 6DOF human motion tracking using miniature inertial sensors," Xsens Technologies B.V., Technical Report, 2009.

[19] M. Schepers, M. Giuberti, and G. Bellusci, "Xsens MVN: Consistent tracking of human motion using inertial sensing," Xsens Technologies B.V., Technical Report, 2018.

[20] NaturalPoint, Inc., "OptiTrack motive documentation: Rigid body tracking," https://docs.optitrack.com/, 2024, accessed Sep. 2026.